



Store at: +2+8°C.

Presentation:

Cod. EZ007 CONT: R1 1 x 40 mL. + R2 x 10 mL.

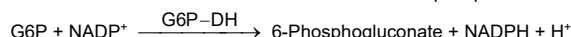
## Procedure

### Quantitative determination of creatine kinase (CK).

#### Only for in vitro use in clinical laboratory (IVD)

#### TEST SUMMARY

Creatine kinase (CK) catalyses the reversible transfer of a phosphate group from phosphocreatine to ADP. This reaction is coupled to those catalysed by hexokinase (HK) and glucose-6-phosphate dehydrogenase (G6P-DH):



The rate of NADPH formation, measured photometrically, is proportional to the catalytic concentration of CK present in the sample<sup>1,2</sup>.

#### REAGENTS COMPOSITION

|                  |                                            |             |
|------------------|--------------------------------------------|-------------|
| R 1<br>Buffer    | Imidazol pH 7.0                            | 125 mmol/L  |
|                  | Glucose                                    | 25 mmol/L   |
|                  | N-acetyl cysteine                          | 25 mmol/L   |
|                  | Magnesium acetate                          | 12.5 mmol/L |
|                  | NADP <sup>+</sup>                          | 2.5 mmol/L  |
|                  | EDTA                                       | 2 mmol/L    |
| R 2<br>Substrate | Hexokinase (HK)                            | ≥6800U/L    |
|                  | Imidazol pH 8,90                           | 125 mmol/L  |
|                  | ADP                                        | 15.2 mmol/L |
|                  | AMP                                        | 25 mmol/L   |
|                  | di-Adenosine-5- pentaphosphate             | 103 mmol/L  |
|                  | Glucose-6-phosphate dehydrogenase (G6P-DH) | ≥8800 U/L   |
|                  | Creatine phosphate                         | 250 mmol/L  |

#### PRECAUTIONS

R1/R2: H360D- May damage the unborn child. Restricted to professional users.

Follow the precautionary statements given in MSDS and label of the product.

#### REAGENT PREPARATION AND STABILITY

Working reagent (WR): Mix 1 volume of R2 with 4 volumes of R1.

Stability: 14 days at 2-8° C or 48 hours at room temperature (15-25° C).

#### Signs of Reagent deterioration:

- Presence of particles and turbidity.
- Blank absorbance (A) at 340 nm. ≥ 1.00

All the components of the kit are stable until the expiration date on the label when stored at 2-8°C, protected from light and contamination prevented during their use. Do not use reagents over the expiration date. Do not freeze.

#### SPECIMEN

Serum free of haemolysis or heparinized plasma<sup>1</sup>: Stability: 2 days at 2-8°C, 8h at room temperature or 1 month at -20°C, protected from light. CK activity decreases 10% after 1 day at 2-8° C. or after 1<sup>h</sup> hour at 15-25° C.

#### MATERIAL REQUIRED BUT NOT PROVIDED

- Spectrophotometer or colorimeter measuring at 340 nm.
- Thermostatic bath at 25° C, 30° C o 37° C (± 0.1° C)
- Matched cuvettes 1.0 cm. light path.

#### General laboratory equipment.

#### TEST PROCEDURE

- Assay Conditions
  - Wavelength: ..... 340 nm.
  - Cuvette: ..... 1 cm light path.
  - Constant temperature ..... 25°C / 30°C / 37°C.
- Adjust the instrument to zero with distilled water or air.
- Pipette into a Cuvette (note 1):

|              | 25-30°C | 37°C. |
|--------------|---------|-------|
| WR (mL)      | 1.0     | 1.0   |
| Sample (μL.) | 40      | 20    |

- Mix and incubate for 2 minutes.
- Read the absorbance (A) of the sample, start the stopwatch and read absorbance at 1 min. interval thereafter for 3 min.
- Calculate the difference of absorbance and the average absorbance difference per minute (ΔA/min.)

#### CALCULATIONS (Note 2)

25 – 30° C ΔA/min x 4127\* = U/L of CK

37° C. ΔA/min x 8095\* = U/L of CK

**Units:** One international unit (IU) is the amount of enzyme that transforms 1 μmol of substrate per minute, in standard conditions. The concentration is expressed in units per litre of sample (U/L).

#### Temperature conversion factors

To correct results to other temperatures multiply by:

| Assay temperature | Conversion factor to |      |      |
|-------------------|----------------------|------|------|
|                   | 25°C                 | 30°C | 37°C |
| 25°C              | 1.00                 | 1.56 | 2.44 |
| 30°C              | 0.64                 | 1.00 | 1.56 |
| 37°C              | 0.41                 | 0.63 | 1.00 |

#### QUALITY CONTROL

Control sera are recommended to monitor the performance of the procedure, Control H Normal Ref. QC003 and Control H Pathological Ref. QC004. If control values are found outside the defined range, check the instrument, reagents and calibrator for problems.

Serum controls are recommended for internal quality control. Each laboratory should establish its own Quality Control scheme and corrective actions.

#### REFERENCE VALUES<sup>1</sup>

|             | 25° C   | 30° C    | 37° C    |
|-------------|---------|----------|----------|
| Men up to   | 80 U/L. | 130 U/L. | 195 U/L. |
| Women up to | 70 U/L. | 110 U/L. | 170 U/L. |

(These values are for orientation purpose).

It is suggested that each laboratory establish its own reference range.

#### CLINICAL SIGNIFICANCE

Creatine kinase is a cellular enzyme with wide tissue distribution in the body. Its physiological role is associated with adenosine triphosphate (ATP) generation for contractile or transport systems.

Elevated CK values are observed in diseases of skeletal muscle and after myocardial infarction<sup>1,5,6</sup>.

Clinical diagnosis should not be made on a single test result; it should integrate clinical and other laboratory data.

#### REAGENT PERFORMANCE

##### Measuring Range:

From detection limit of 10 U/L. to linearity limit of 1714 U/L., under the described assay conditions.

If results obtained were greater than linearity limit, dilute the sample 1/10 with NaCl 9 g/L. and multiply result by 10.

##### Precision:

|            | Intra-assay n= 20 |      | Inter-assay n= 20 |      |
|------------|-------------------|------|-------------------|------|
| Mean (U/L) | 147               | 494  | 145               | 485  |
| SD         | 1.23              | 3.60 | 2.91              | 8.97 |
| CV (%)     | 0.84              | 0.73 | 2.01              | 1.85 |

##### Sensitivity: 1 U/L

**Accuracy:** Results obtained reagents did not show systematic differences when compared with other commercial reagents.

Correlation coefficient (r)<sup>2</sup>: 0.9995.

Regression Equation: y= 1.0846x – 0.3512

The results of the performance characteristics depend on the analyzer used.

#### INTERFERING SUBSTANCES

- No interferences were observed with glucose up to < 7 g/L, triglycerides up to 7 mmol/L and haemoglobin up to 5 g/L<sup>1,2</sup>
- A list of drugs and other interfering substances with CK determination has been reported by Young et. Al<sup>3,4</sup>.

#### NOTES

- Use clean disposable pipette tips for its dispensation.
- Formulation to reach constant:

ΔA/min x 4127\* or  
8095\* = U/L CK

|                            |                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------|
| * Tv x 1000<br>ε x LP x Sv | Tv= Total volume in mL<br>ε NADPH = 6.22 at 340 nm<br>LP= Light path<br>Sv= Sample volume in mL |
|----------------------------|-------------------------------------------------------------------------------------------------|

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